Experiment No. 5

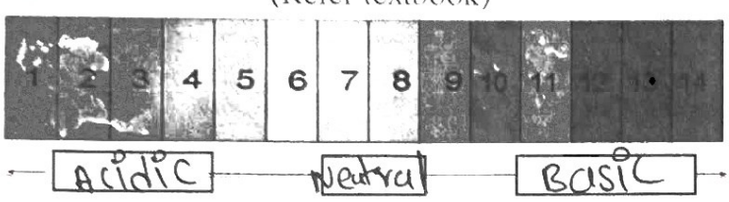
Title : To determine the pH of the given samples using pH paper or Universal indicator.

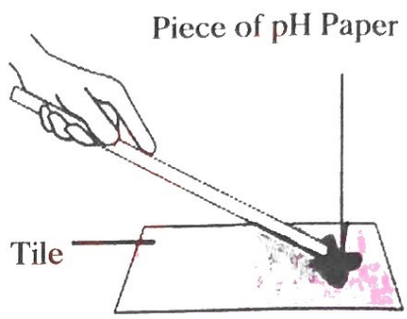
Apparatus : Test tubes, Test tube stand, Droper, White glazed tile, pH paper, Universal Indicator, Distilled water etc.

Given Sample : Dilute Hydrochloric acid, Solution of dilute Sodium Hydroxide, Dilute Acetic acid, Lemon Juice, Dilute Sodium Bicarbonate Solution etc.

Figure :

(Refer textbook)





Procedure :

A. By using pH paper.

1. Place six clean and dry test tubes in a test tube stand.
2. Take 10 ml each of the solution in the test tubes.
3. Place a small piece of pH paper with the help of dropper on a glazed tile.
4. Place one or two drops of the solution in the first test tube on the pH paper with the help of a dropper.
5. Compare the colour produced on the pH paper and the colours in the chart on the pH paper.
6. Determine the pH of solution from the colour produced on the pH paper and note it.
7. Repeat the same procedure for each of the given solution and note the pH of each of the solutions in the observation table.

Observation Table :

Sr.No	Name of the Solution	The colour produced	pH
1	dilute hydrochloric acid	Reddish	1
2	dilute Sodium hydroxide	Blue	12
3	dilute ethanoic acid	orange	3
4	lemon juice	orange	3
5	water	No change light Green.	7
6	dilute sodium bicarbonate	voilet.	10

B. By using Universal indicator

1. Place six clean and dry test tubes in test tube stand.
2. Take 10 ml each of the solution in the test tubes.
3. Put two-two drops of universal indicator in each of the test tubes with the help of dropper.
4. Observe the colour change in the solution in each test tube.
5. Compare with the colour shades on the Universal indicator bottle.
6. From that determine the pH of the solution and note it in the observation table.

Observation table :

Sr.No	Name of the Solution	The colour produced	pH
1	dilute hydrochloric acid	Raddish	1
2	dilute Sodium hydroxide	Blue	12
3	dilute ethanoic acid	orange	3
4	lemon juice	orange	3
5	water	no change / light Green	7
6	dilute sodium bicarbonate	voilet.	10

Inference :

1. Acidic Solutions (pH less than 7). Dilute hydrochloric acid dilute ethonic.
2. Basic Solutions (pH more than 7). Dilute sodium hydrocloric dilute sodium.
3. Neutral Solutions (pH = 7). 1 cocter.

Multiple Choice Questions

1. Some aqueous soluions have the following pH values. Which is the acidic solution?
   A. pH = 6 B. pH = 7 C. pH = 10 D. pH = 14
2. Colour of the Universal indicator paper is .....
   A. Red B. Blue C. Green D. Greenish Yellow
3. Which of the following solution has pH greater than 7?
   A. Pure water B. NaOH C. HCl D. lemon juice
4. Which of the following solution has pH 7 ?
   A. Pure Water B. aqueous solution of H_2SO_4
   C. aqueous solution of Na_2CO_3 D. aqueous solution of HCl

Questions

1. Explain the importance of pH testing in everyday life.

P.H plays a very important role in our everyday life. In our digestive system hydrocloric acid produced in our stomach helps the digestive of Road without causing and harm to the stomach.

2. How will you corelate the measures used in agriculture with acids and bases?

1) There are somany types of soil some and acidic & some are bassic.
2) Neutral soil is nessasary for plants because of use of fertilizers.

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